

A075 – Field Theory: Mode Functions, Lagrangian, and Bridge Formula

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.1 Purpose

[STIPULATION] This document develops the complete field theory of FFGFT: explicit mode functions on the 4-torus, the free Lagrangian, the bridge formula between Lagrangian and mode operator, the Dirac and Schrödinger equations as low-energy limits, the binding picture of masses, and the non-closure theorem. It is the link between the geometric structure (A030/A040) and the sectors (A100 ff.) – it shows that masses do not appear as parameters in the Lagrangian but emerge as eigenvalues of the mode operator.

.2 The Basic Geometry of the Torus

[B] The fundamental carrier is the compact 4-torus T^4 with the characteristic length scale

$$L_T = \frac{\ell_P}{\xi} \approx 7500 \ell_P \approx 6.14 \times 10^{-16} \text{ GeV}^{-1}.$$

(Not to be confused with the minimal length scale $L_0 = \xi \cdot \ell_P \approx 2.15 \times 10^{-39} \text{ m}$ from A015.) This is the *fundamental period* of the torus: all particles are modes on L_T – like overtones of a vibrating string in three spatial dimensions. The Compton wavelength $L_i = 2\pi/m_i$ is the wavelength of the respective mode, not the size of a separate torus.

[B] The FFGFT torus carries a fractal correction of the dimension:

$$D_f = 3 - \xi \approx 2.9998667.$$

This minimal deviation is the origin of all particle masses. It generates a non-linear dispersion relation – analogous to an anharmonic crystal lattice. On an exactly three-dimensional torus, masses would be integer-proportional to the wave number; the observed ratios $m_\mu/m_e \approx 206.8$ and $m_\tau/m_e \approx 3477$ force the fractal dispersion.

.3 Quantum Numbers of the Modes

[B] Each mode on the torus carries three quantum numbers: n (principal quantum number), l (orbital angular momentum, transverse component), j (spin quantum number, $j = l \pm \frac{1}{2}$ for fermions). The wave vector is quantised:

$$\mathbf{k}_{n,l} = \frac{2\pi}{L_0}(k_x, k_y, k_z), \quad k_x, k_y, k_z \in \mathbb{Z},$$

with $n = k_x^2 + k_y^2 + k_z^2$ and $l = \sqrt{k_x^2 + k_y^2}$. The assignment for the leptons:

Particle	n	l	(k_x, k_y, k_z)	r_i	p_i
Electron	1	0	(1, 0, 0)	4/3	3/2
Muon	2	1	(1, 1, 0)	16/5	1
Tau	3	2	(1, 1, 1)	25/9	2/3

(Note of 30 Sept. 2026: By the document's own definition $l = \sqrt{k_x^2 + k_y^2}$, the vectors (1, 0, 0), (1, 1, 0), (1, 1, 1) give $l = 1, \sqrt{2}, \sqrt{2}$ (and non-integer), not 0, 1, 2; the definition and the l column contradict each other.)

[K] The lepton exponents $(p_e, p_\mu, p_\tau) = (3/2, 1, 2/3)$ form a geometric sequence with ratio $2/3$ – the inverse of the tetrahedral factor $4/3$ from $\xi = 4/3 \times 10^{-4}$:

$$p_{n+1} = \frac{2}{3} p_n.$$

Up-type quarks (up to Charm) follow the same scheme with a step skipped; Down-type quarks an arithmetic sequence ($\Delta p = -1/2$); the top quark ($p_t = -1/3, r_t = 1/28$) breaks the scheme – no known geometric reason (open, A150).

.4 Explicit Mode Functions

[B] The eigenfunctions of the d'Alembertian $\square = \partial_t^2 - \nabla^2$ on $T^3 \times \mathbb{R}$ are:

$$\phi_{n,l,j}(t, \mathbf{x}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_{n,l} \cdot \mathbf{x}} \cdot Y_{l,j}(\vartheta, \varphi) \cdot e^{-im_{n,l,j}t}$$

with torus volume $V = L_0^3$ and spherical harmonics $Y_{l,j}$. The modes are orthonormal:

$$\int_{T^3} \phi_{n,l,j}^*(\mathbf{x}) \phi_{n',l',j'}(\mathbf{x}) d^3x = \delta_{nn'} \delta_{ll'} \delta_{jj'}$$

and satisfy the Klein–Gordon equation:

$$\square \phi_{n,l,j} = -m_{n,l,j}^2 \phi_{n,l,j}.$$

The spin structure follows from the winding numbers of the torus: spin- $\frac{1}{2}$ corresponds to the winding $(n_\theta, n_\phi) = (1, 1)$; the 720° symmetry follows directly from the topology.

.5 The Lagrangian

[B] The starting point is the free Klein–Gordon Lagrangian:

$$\mathcal{L}_0 = \frac{1}{2}(\partial \delta m)^2,$$

the scalar field $\delta m(x)$ describes mass fluctuations on the torus; the equation of motion $\square \delta m = 0$ describes massless oscillations. The fractal geometry adds a correction term generating the non-linear dispersion:

$$\Delta \mathcal{L}_{\text{Torus}} = \varepsilon \xi \left[\frac{f(\vartheta)}{2} (\partial \delta m)^2 - k^{\mu\nu}(\vartheta) \partial_\mu \delta m \partial_\nu \delta m \right].$$

The geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ encode the curvature of the fractal torus. The complete Lagrangian:

$$\mathcal{L} = \mathcal{L}_0 + \Delta \mathcal{L}_{\text{Torus}}$$

(Note of 2 Oct. 2026 on notation: here $f(\vartheta)$ is a geometric function in the torus Lagrangian and $f(k_t)$ in $m_t c^2 = \hbar f(k_t)$ the winding-energy function, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

[S] Connection to the Standard Model. For $\xi \rightarrow 0$, $\Delta \mathcal{L}_{\text{Torus}}$ vanishes; FPGFT contains the SM Lagrangian as a limiting case. The difference from Yukawa: in the SM the coupling y_t is a free parameter; in FPGFT the corresponding term $\xi \cdot m_t$ is geometrically fixed. (Note of 2 Oct. 2026: the limit $\xi \rightarrow 0$ is shown only for the scalar field δm : $\Delta \mathcal{L}_{\text{Torus}}$ vanishes, $\mathcal{L}_0$ remains. Gauge, fermion and Higgs terms appear in A142 coupled to the time field; that their limit yields the full SM Lagrangian is not calculated there. The statement is therefore a setting, not a numerical finding; previously listed as [K].)

.6 The Mode Operator

[B] The Hilbert space of torus modes:

$$\mathcal{H} = \bigoplus_{n,l,j} \mathbb{C} \cdot \mathbf{e}_{n,l,j}, \quad \langle \mathbf{e}_{n,l,j}, \mathbf{e}_{n',l',j'} \rangle = \delta_{nn'} \delta_{ll'} \delta_{jj'}.$$

The field operator:

$$\Phi(x) = \sum_{n,l,j} m_{n,l,j} \cdot \phi_{n,l,j}(x) \cdot P_{n,l,j}$$

with projectors $P_i = |\mathbf{e}_i\rangle\langle\mathbf{e}_i|$. Since $P_i P_j = \delta_{ij} P_i$, we have $\Phi^2 = \sum_i m_i^2 P_i$.

.7 The Bridge Formula: Lagrangian and Mode Operator

[B] The core of the field theory: the Lagrangian and the mode operator describe the same physics. The bridge formula in six steps:

$$\int \frac{1}{2} (\partial \delta m)^2 d^4 x = \frac{1}{2} \langle \psi | \Phi^2 | \psi \rangle$$

Step 1 – Mode expansion: $\delta m(x) = \sum_i a_i \phi_i(x)$.

Step 2 – Substitution: $\int \mathcal{L}_0 d^4 x = \frac{1}{2} \sum_{i,j} a_i a_j \int (\partial \phi_i)(\partial \phi_j) d^4 x$.

Step 3 – Orthogonality: $\int (\partial \phi_i)(\partial \phi_j) d^4 x = 0$ for $i \neq j$. Hence $= \frac{1}{2} \sum_i a_i^2 \int (\partial \phi_i)^2 d^4 x$.

Step 4 – Klein–Gordon (partial integration): $\int (\partial \phi_i)^2 d^4 x = m_i^2$ (normalisation).

Step 5 – Summary: $\int \mathcal{L}_0 d^4x = \frac{1}{2} \sum_i a_i^2 m_i^2$.

Step 6 – Mode operator: $\langle \psi | \Phi^2 | \psi \rangle = \sum_i a_i^2 m_i^2$. Comparison yields the bridge formula.

[B] Three description levels are thereby equivalent:

Level	Formula
Lagrangian	$\mathcal{L}_0 = \frac{1}{2}(\partial \delta m)^2$
Mode operator	$\Phi = \sum_i m_i P_i$
Bridge	$\int \mathcal{L}_0 d^4x = \frac{1}{2} \langle \psi \Phi^2 \psi \rangle$

Mass arises not as a Lagrangian parameter but as an eigenvalue of the mode operator – this is the structural difference from the Standard Model.

.8 Mass Generation as a Binding Phenomenon

[B] The eigenvalue equation on the torus:

$$[\square + \hat{V}(\xi, D_f)] \phi_{n,l,j} = -m_{n,l,j}^2 \phi_{n,l,j}.$$

The mass results as the difference of kinetic energy and potential:

$$m_i^2 = \underbrace{\left(\frac{2\pi n}{L_0} \right)^2}_{\text{kinetic}} + \underbrace{\langle \phi_i | \hat{V} | \phi_i \rangle}_{\approx -(2\pi n/L_0)^2}.$$

(Note of 30 Sept. 2026: The numerical values of the kinetic term ($1.046, 4.185, 9.415 \times 10^{32} \text{ GeV}^2$ for e, μ, τ) follow only with $L_T = \ell_P/\xi = 6.14 \times 10^{-16} \text{ GeV}^{-1}$, not with the $L_0 = \xi \ell_P$ in the formula (which would give $(2\pi/L_0)^2 \approx 3.3 \times 10^{47} \text{ GeV}^2$). Moreover $|\mathbf{k}|^2 = (2\pi/L)^2 n$ is linear in $n = k_x^2 + k_y^2 + k_z^2$, whereas $(2\pi n/L)^2$ scales with n^2 ($1 : 4 : 9$). Linear in n gives $1.046, 2.092, 3.138 \times 10^{32} \text{ GeV}^2$.)

[S] The kinetic energy is enormous ($\sim 10^{32} \text{ GeV}^2$), the potential almost exactly opposite. The mass is the tiny remainder. This is not fine tuning but the defining property of bound states. *Crystal analogy:* one does not ask why a crystal produces a particular phonon dispersion – the dispersion curve is the crystal structure in momentum space. *Hydrogen analogy:* kinetic energy $\approx 13.6 \text{ eV}$, Coulomb potential $\approx -27.2 \text{ eV}$, binding energy -13.6 eV – no one asks why the potential compensates the kinetic energy so precisely, that follows from the Schrödinger equation. In FFGFT: the particles are bound torus modes; the potential $\hat{V}$ is uniquely determined by ξ and D_f – no free parameter.

.9 The Non-Closure Theorem

[K] For all pairs of distinct torus modes $i \neq j$:

$$\forall i \neq j : (r_i \cdot r_j, p_i + p_j) \notin \mathcal{P},$$

where $\mathcal{P}$ is the set of all pairs (r, p) admitted in FFGFT.

Example: $r_e \cdot r_\mu = \frac{4}{3} \cdot \frac{16}{5} = \frac{64}{15}$, $p_e + p_\mu = \frac{3}{2} + 1 = \frac{5}{2}$. In the complete mode table there is no particle with $r = 64/15$ and $p = 5/2$.

[B] Why does the theorem hold? The cause lies in the orthogonality of the modes: in Hilbert space the product of two distinct basis vectors is structurally meaningless – a category violation. $r_e \cdot r_\mu$ is computable as a number, but belongs to no mode of the torus.

[K] Practical rule: *Permitted* is numerical evaluation ($m_e \cdot m_\mu \approx 53.9 \text{ MeV}^2$) and the mass formula $m_i = r_i \cdot \xi^{p_i} \cdot v$ for fixed i . *Forbidden* is symbolic cross-multiplication: $(r_i \cdot r_j) \cdot \xi^{p_i+p_j} \cdot v$ for $i \neq j$.

.10 Dirac Line and Schrödinger Limit

[B] Within the Lagrangian framework the geometric equation replaces the full Dirac equation by:

$$\partial^2 \Delta m = 0.$$

The fundamental structure is the Clifford algebra $\mathbf{e}_\mu \mathbf{e}_\nu + \mathbf{e}_\nu \mathbf{e}_\mu = 2g_{\mu\nu}$; the 4×4 γ -matrices are retained as a representation of the Clifford generators for concrete computations – they are not eliminated but traced back to their geometric meaning. Spin manifests on the fractal torus as a topological property (vacuum-phase winding). A relativistic state:

$$\Phi(x, t) = \rho(x, t) e^{i\theta(x, t)} \cdot \chi(\sigma),$$

with spin component $\chi(\sigma)$ from the geometric knot modes.

[B] In the non-relativistic limit, the phase equation of the vacuum field reduces to the standard Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi,$$

with $\psi \propto \sqrt{\rho} e^{i\theta}$, effective mass m from the knot inertia, and potential V from external ρ -perturbations. The Schrödinger equation is *not fundamental* but an effective equation for slow, non-relativistic knot excitations – the low-energy limit of the FFGFT mode structure. This limit follows from the Lagrangian framework via the DVFT polar decomposition $\Phi \rightarrow \sqrt{\rho} e^{i\theta}$.

[B] The transition Lagrangian $\rightarrow$ Schrödinger equation in three stages: (1) mode superposition in the low-energy limit: $\Phi(x) \xrightarrow{\text{low energy}} \sqrt{\rho(x, t)} e^{i\theta(x, t)}$ with energy density ρ and coherent phase θ . (2) The phase dynamics of θ satisfies the Schrödinger equation in the non-relativistic limit. (3) Masses follow from k_t -windings via $m_i c^2 = \hbar f(k_t^{(i)})$; the lepton exponents $(3/2, 1, 2/3)$ are the geometric sequence of this tower.

.11 Algebras, Recursion Operator, and Scale Structure

[B] The recursion operator $\mathcal{R}$ formalises the fractal scaling structure, diagonal on the mode space:

$$\mathcal{R}(\mathbf{e}_{n,l,j}) = \xi^{p(n,l,j)-1} \cdot K_{\text{frak}} \cdot \mathbf{e}_{n,l,j}.$$

The time-mass duality is encoded as a field symmetry: $\mathcal{R} : t \rightarrow \xi t, m \rightarrow \xi^{-1} m, \Phi \rightarrow \Phi$. The mode algebra for free waves: $\mathbf{e}_{\mathbf{k}_i} \star \mathbf{e}_{\mathbf{k}_j} = \mathbf{e}_{\mathbf{k}_i + \mathbf{k}_j}$, with structure constants $C_{ij}^k = \delta_{k,i+j \bmod \text{torus period}}$.

[K] The scale running of the coupling – in FFGFT-native language: an $\mathcal{R}$ -induced scale correction, not a renormalisation procedure – gives in the QED analogy:

$$\alpha^{-1}(\mu) = \alpha^{-1}(m_e) - \frac{\beta_0}{2\pi} \ln \frac{\mu}{m_e} - \frac{\beta_0 \xi}{4\pi} \left(\ln \frac{\mu}{m_e} \right)^2.$$

The ξ -term is the FFGFT-specific addition; ξ is not a fixed point of the β -function but a geometrically fixed constant (A020).

.12 Methodological Position of the Field Theory

[K] Reduction is not increased precision. Where the Standard Model is experimentally confirmed (QED scattering amplitudes to $\sim 10^{-12}$), FFGFT can only reproduce these plus ξ -corrections of order 10^{-4} , not qualitatively surpass them. The FFGFT mass formulas give values at $\sim 1\%$ agreement with PDG values – that is a *structural proof* (masses follow from geometry, not free parameters), not a precision calculation. The reduction from ~ 25 free SM parameters to a single geometric parameter ξ is the quantifiable statement.

[S] Circularity of the comparison basis. PDG values are not model-free raw data; they contain assumptions about units, gauge conventions, and SI-2019-reform definitions. An FFGFT calculation in natural units ($\hbar = c = 1$) avoids the accumulation of model assumptions; as soon as it is compared against PDG data, it inherits their prior corrections. The $\sim 1\%$ comparison is therefore to be read in two directions: confirmation of the geometric structure *and* indication of the latitude of the circular measurement basis.

.13 Open Mathematical Tasks

The Hopf algebra for interactions is not worked out. The free theory has structure constants $C_{ij}^k = \delta_{k,i+j}$ (wave vector addition). Complete Feynman rules from the structure constants and the non-trivial fixed point of the β -function (whether μ^* exists and what it means physically) are open.

Top quark, Down-type step size, quark r_i deep structure. For leptons and Up-type quarks (up to Charm) the geometric $2/3$ -sequence of exponents is known; for the Top ($p_t = -1/3$) and Down-type (arithmetic sequence $\Delta p = -1/2$) the geometric justification is missing. The $\alpha\text{-}\pi\text{-}\sqrt{3}$ deep structure of the lepton r_i is partially present (A130); the analogue for quark r_i is missing (A150).

The test condition is structural, not quantitative. The test $\langle \phi_i | \hat{V} | \phi_i \rangle = r_i^2 \xi^{2p_i} v^2 - (2\pi n/L_0)^2$ is the bottleneck: explicitly deriving $\Delta \mathcal{L}_{\text{Torus}}$ from the torus geometry and showing that the eigenvalues agree with the mass table – as phonon dispersion from lattice forces of a crystal.

.14 Verification Scripts

- Mode functions and orthonormality: verification $\int \phi_i^* \phi_j d^3x = \delta_{ij}$ for $i, j \in \{e, \mu, \tau\}$ numerically: `python/A_Serie_Skripte/a075_modefunktionen.py`
- Bridge formula: $\int \frac{1}{2}(\partial \delta m)^2 d^4x = \frac{1}{2} \langle \psi | \Phi^2 | \psi \rangle$ for random coefficients (a_i) : `python/A_Serie_Skripte/a075_bruecke.py`
- Non-closure theorem: verification that $(r_e r_\mu, p_e + p_\mu)$ does not lie in $\mathcal{P}$: `python/A_Serie_Skripte/a075_nichtabschluss.py`

.15 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\square = \partial_t^2 - \nabla^2$	d'Alembertian (wave operator)
$\mathcal{L}$	Lagrangian density
$\phi_{n,l,j}(x)$	Mode function on T^4 ; quantum numbers as in A100
$m_{n,l,j}$	Mass of mode (n, l, j)
Ψ_ℓ	Lepton spinor field
$g_T^\ell = m_\ell \xi$	Mass-proportional coupling of the time field

.16 Open Questions

The complete form of $\Delta\mathcal{L}_{\text{Torus}}$ – the geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ – is not derived in closed form from the torus geometry. The correction factor $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ is known (A040), the complete tensor function is not.

The quark-sector extension of the mode table and the non-closure theorem to all twelve fermions is worked out for leptons and Up-type quarks (up to Charm), but not for Down-type and Top (A150).

The Hopf algebra structure for interactions and complete Feynman rules are outstanding.

.17 Supersedes

This document subsumes: Doc. 202 (complete FFGFT field theory: fundamental scale L_0 , quantum numbers of modes, explicit mode functions $\phi_{n,l,j}$, free Lagrangian $\mathcal{L}_0$, torus correction term, mode operator $\Phi = \sum m_i P_i$, bridge formula in six steps, non-closure theorem, binding picture of masses, Dirac and Schrödinger limits, recursion operator $\mathcal{R}$, scale structure). *Not taken over*: the large-scale sector from Doc. 202 (DVFT line on vacuum field effects at galactic scales) – that does not belong in the A-series core (A010); the section on circularity of the maximal-formula notation is left to the correction programme.

.18 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.